

Investigating the Effectiveness of Pain Masking Through Contextual Modification

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Abstract

Drug administration is known to effectively alleviate pain. However, due to the side effects of many drugs, there is a growing need for alternative, non-pharmacological pain relief methods. This study proposes a method that misidentifies the cause of pain and reduces its intensity by providing contextual information through video content. For instance, when a cute kitten appears in a video alongside a "scratching" action and pain, the kitten's action is perceived as the cause of the pain, transforming the experience into a pleasant interaction. To evaluate the effectiveness of this approach, the study examined whether the interpretation of pain stimulation, the level of discomfort, and the pain tolerance threshold were affected when electrical stimulation from the side of a smartphone was used as a source of pain and a cat scratching animation was displayed on the screen under varying conditions. The results indicated that participants with a higher tolerance for electrical stimulation experienced an increase in their tolerance for discomfort when the visual and sensory stimuli were presented together. Additionally, participants were more likely to attribute the stimulation to the cat's action. However, no significant differences were observed in responses to pain-related questions. These findings suggest that while this method may not reduce the intensity of pain, it could lower the discomfort associated with it.

CCS Concepts

• **Human-centered computing** → **Visualization toolkits**; **Visualization application domains**; **Haptic devices**.

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1 Introduction

The administration of drugs such as opioids and anesthetics effectively alleviates pain [10]. However, these drugs are linked to various side effects. For instance, opioids can lead to dizziness, nausea, and physical dependence [2]. Many individuals receiving pain relief therapy express discomfort with drug-based treatments due to these side effects. As a result, there is a growing need to develop non-pharmacological methods of pain management that do not involve anesthetics or analgesics.

One method of reducing pain without medication is to distract attention, thereby reducing the perception of pain [13, 19]. In one study, children who played with cards or kaleidoscopes while blood was being drawn reported lower levels of pain during the procedure [3]. A study of adolescents also indicated that they were less likely to report pain when distracted by music through headphones during vaccinations [11]. These distraction-based methods for pain reduction have been examined in medical settings [5, 6, 8, 15, 18] and among children [1, 4, 7, 17, 22].

An alternative approach to distraction involves modifying the appearance of the virtual arm in virtual reality (VR) space. This change influenced the subject's response, measured by skin conductance, when the virtual needle approached or made contact [12]. In a study of patients with chronic hand pain, pain was reduced when patients performed 10 standardized hand movements, with the movements displayed to them at a smaller size than their actual size [16].

Most non-drug methods of pain reduction, as described earlier, focus on "distracting the patient from the pain" or "imagining a less painful self." However, these distraction methods are less effective for severe pain [14] (the pain is "surprising", if the patients were distracted), and it is challenging to distract individuals who are highly fearful of pain[9].

This study introduces a new method of pain reduction called "context modification," as shown in Figure 1 Context modification involves altering the perceived cause of sensation through animation presented in virtual reality (VR) space or on a smartphone. For instance, if a kitten appears in front of the user's eyes using augmented reality technology, and the kitten "scratches" the user while pain is also presented, the pain is no longer experienced as pain, but as a pleasant interaction. In contrast to traditional "pain distraction" methods, the individual receiving the pain stimulus directs attention toward the stimulus itself.

This study investigates whether displaying an animation of a kitten on a smartphone screen reduces the pain felt by a grasping hand. Compared to other pain induction methods, such as heat, electrical stimulation can produce a pain-like sensation [23], and its intensity can be adjusted in real time.

This study is inspired by the work of Takami [20] et al., in which electrical stimulation from electrodes placed on the side of a smartphone was synchronized with an animation of a kitten on the smartphone screen to induce the sensation of "playing with a cat." Users who saw the animation sought more intense stimulation, and their perception of the sensory quality of the electrical stimulation became more favorable. The present study further investigates the effects of contextual modification on pain perception.

This study introduces the concept of "pain masking by contextual modification," which seeks to reduce the perception of pain through the misidentification of its cause. The key contributions of this research are as follows:

- The study confirmed the method's effectiveness by demonstrating that misidentifying the source of pain can reduce its perceived intensity.
- It examined the applicability of this approach to individuals with varying levels of tolerance to electrical pain stimulation, including those unable to endure it.



Figure 1: Proposed method: A new context is established by presenting both a stimulus and its corresponding cause at the same time.

2 Experiment 1:(Examining the Threshold of Electrical Stimulation at the Point of Discomfort Through Contextual Modification)

This study investigated whether contextual modification increases stimulation to the threshold of discomfort. Twenty-one participants (19 males and 2 females, mean age 28.4 years) were involved in the experiment. All experimental procedures were reviewed and approved by the ethics committee of the authors' institution.

2.1 Device

This experiment utilized a smartphone integrated with a side-mounted electrical stimulation device, as illustrated in Figure 2(a). The setup was based on the system developed by Takami et al. for interaction with cats [20]. The smartphone was equipped with 32 electrodes on both sides, each with a diameter of 2.0 mm and a center-to-center distance of 3.75 mm. The device could generate a maximum current of 23 mA and a maximum pulse duration of 1 ms. For this study, the maximum current was set to 10 mA and a pulse width was fixed to 50 μ s. Only three electrodes depicted in Figure 2(a), located where the middle finger contacted the device, were used; the remaining electrodes were covered with masking tape.

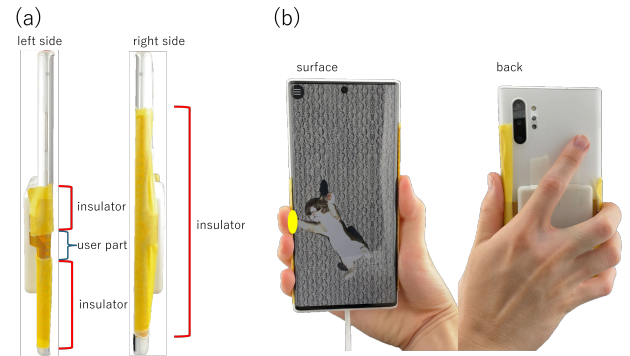


Figure 2: (a) An electrode positioned on the side of the smartphone; (b) A cat image displayed on both the screen and the back of the smartphone during the experiment.

2.2 Experimental Procedure

Subjects were instructed beforehand on the proper handling of the experimental device. Each subject was seated in a chair and held the device with their right hand. As illustrated in Figure 2(b), the little, ring, and middle fingers were placed on the left side of the smartphone, the thumb was positioned on the right side, and the index finger contacted the back of the device. The middle finger was positioned so that the finger pad touched and fully covered the side electrode of the smartphone. To maintain consistent positioning, we marked the center of the finger with a marker.

In this condition, electrical stimulation was administered at regular intervals to the middle finger of the right hand. The intensity of the stimulation was progressively increased, starting at 0 mA. The threshold current value was recorded when the participant

indicated that the stimulation had become uncomfortable. The stimulation was immediately ceased by the experimenter upon the onset of discomfort. Three experimental conditions were conducted in a randomized sequence for each participant, with intervals of several dozen seconds between conditions to eliminate any residual effects of prior stimulation.

The experimental data were analyzed using the Friedman test to assess statistical differences across conditions. When significant differences were detected, post hoc analyses were performed using the Wilcoxon signed-rank test with Bonferroni correction for each condition.

2.3 Conditions

Three experimental conditions were designed, as depicted in Figure 3. Condition 1 involved displaying a background image while applying electrical stimulation at regular intervals to the middle finger of the participant's right hand. Condition 2 introduced an animated 3D model of a cat on the screen, with identical electrical stimulation delivered when the cat appeared to scratch the user's hand, as illustrated in Figure 2(b). Condition 3 introduced a 0.45-second delay in the timing of the electrical stimulation relative to condition 2, while all other parameters remained unchanged. These conditions were termed "nothing," "cat," and "lag," respectively. The inclusion of the lag condition aimed to explore the effect of temporal delays between modalities. The delay duration of 0.45 seconds was chosen because it exceeds the perceptual threshold of 0.02 seconds [21] and was identified as perceptible by participants in preliminary tests comparing conditions with and without the lag.

The electrical stimulation parameters remained consistent across all conditions. Anodic stimulation was used, with the center electrode functioning as the anode and two adjacent electrodes functioning as the ground. The stimulation parameters included a pulse width of 50 μ s and a pulse frequency of 60 pulses per second (pps). Animation and stimulation sequences were repeated every 1.97 seconds, corresponding to the duration of the animation loop. Electrical stimulation began at 0 mA and increased incrementally by 0.1 mA per cycle, reaching a peak of 10 mA at the 100th cycle.

The experiment was conducted once for each of the three conditions, presented in a pseudorandom order. Due to the nature of this experiment, which cause discomfort, we limited the number of trials to minimize their burden. A break lasting several tens of seconds was given between conditions to ensure separation.

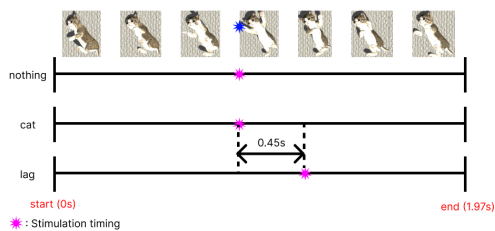


Figure 3: The timing diagram delineates the coordination between electrical stimulation and animation for each experimental condition.

2.4 Results and Discussion

The recorded values of electrical stimulation for each condition were compared. Subjects who were not looking at the smartphone screen during the experiment were excluded from the data. As shown in Figure 4(a), the Friedman test revealed no significant differences. Consequently, no significant differences were observed between the conditions across all subjects.

Figure 4(b) shows a significant variation in response values between subjects. The distribution is bimodal, not normal. Therefore, subjects were divided into two groups based on the electrical stimulation value at "nothing" condition. One group, higher than 4 mA, exhibited high tolerance, while the other exhibited low tolerance.

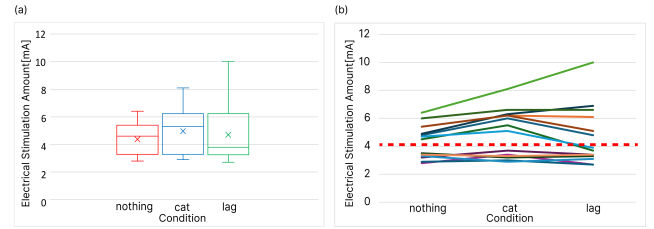


Figure 4: (a) Results from Experiment 1. The values indicate the threshold value of discomfort for each condition. (b) Response values corresponding to each condition and subject, exhibiting a bimodal distribution separated at 4 mA.

The amount of electrical stimulation was compared across conditions in the group with the highest tolerance for electrical stimulation (Figure 5(a)). Friedman and Wilcoxon signed-rank tests revealed a statistically significant difference between the "nothing" and "cat" conditions ($p < 0.05$), while no significant differences were observed between the other conditions. These findings suggest that presenting the cat animation in synchronization with the electrical stimulation timing increased the discomfort threshold, whereas a delay in the animation reduced this effect.

The amount of electrical stimulation was compared across conditions in the low electrical stimulation tolerance group (Figure 5(b)). No significant differences were found between conditions. These results suggest that context modification may not effectively increase the pain threshold for individuals with low tolerance for electrical stimulation.

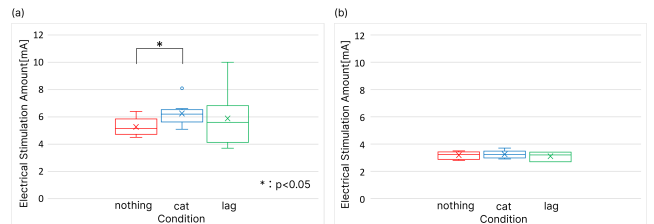


Figure 5: (a) Findings for the group with high tolerance to electrical stimulation. (b) Findings for the group with low tolerance to electrical stimulation.

The results of Experiment 1 suggest that context modification may have increased discomfort thresholds in individuals with a high

tolerance for electrical stimulation. Although it is possible that the increase in discomfort thresholds was due to distraction from the cat image in the experimental setup, the significant differences between the nothing-cat condition and the nothing-lag condition, and the lack of significant differences between the nothing-lag conditions, suggest that the proposed method, "contextual presentation of stimuli," was effective in increasing pain tolerance. Subjects reported that the increase in electrical stimulation was less intense in the cat condition compared to the other conditions. This observation supports the contribution of our method, as well as the conventional method, since subjects did not perceive detailed changes in the stimulation when distracted from its cause.

In the present experiment, the threshold increase was as expected for participants with high tolerance to electrical stimulation, but not for those with low tolerance. Several factors could account for this outcome. The first involves the experimental setup. Due to the setup, individuals with low tolerance were exposed to electrical stimulation for a shorter duration than those with high tolerance (the electrical current value was increased constantly). If a certain amount of exposure time is necessary for context modification, subjects with low tolerance may have reached the threshold before the modification occurred. A second factor concerns the interpretation of the stimulus in individuals with low stimulation tolerance. Those with low tolerance are more likely to experience pain from any level of electrical stimulation, unlike individuals with high tolerance. This may cause the intensity of the stimulation to surpass that of the cat's scratching stimulus. Consequently, the effect of the context modification was diminished because the stimulus was difficult for participants to interpret as a cat scratch. The relationship between stimulation intensity and the observed effect suggests a trend similar to that seen with the distraction method. Moreover, several participants with low tolerance for electrical stimulation noted that they focused more on the stimulation itself than on the cat image. This suggests that when the intensity of the stimulation is too high, participants may focus solely on the stimulation's source.

The variance was larger in the lag condition. This likely occurred because some participants were able to perceive a delay in the lag condition, while others were not. Participants who performed the cat and lag conditions consecutively were more likely to detect the presence or absence of the delay, as they had just completed one condition before the other. In contrast, participants who performed the "nothing" condition between the two conditions were less likely to perceive the delay. Due to the short cycle of the cat animation and the high frequency of stimulation, some participants could not perceive the delay as clearly as others. Subjects who could perceive the delay in the lag condition had thresholds similar to those of the "nothing" condition without context modification, while those who could not perceive the delay had thresholds similar to those of the "cat" and "lag" conditions.

3 Experiment 2 : (Examines How Stimulation was Interpreted)

In Experiment 1, subjects were asked to report the amount of stimulation they found unpleasant. However, pain and discomfort are distinct concepts. For example, some subjects experienced pain as

a sensation but did not report discomfort when presented with the cat image, while others responded solely to the sensation of pain.

To address this issue, this experiment asked participants to report both pain and discomfort. Additionally, the subjects were asked to rate the intensity of sensation produced, whereas Experiment 1 only recorded threshold values. The subjects included 13 participants (11 male and 2 female, average age 22.5 years), 12 of whom had also participated in Experiment 1. The same devices were used as in Experiment 1.

3.1 Experimental Procedure

In this experiment, several tasks were conducted prior to proceeding to the main experiment. Subjects first rated their liking of the cat using a 7-point Likert scale, with 1 representing no liking and 7 representing strong liking. Following this, electrical stimulation was applied to the right middle finger at regular intervals, increasing by 0.1 mA every 1.97 seconds, starting from 0 mA, while only the background image was shown on the screen. The amount of stimulation was recorded when subjects reported discomfort. This recorded level of stimulation served as the criterion for stimulation in the experiment.

The experiment comprised three conditions. The settings were consistent with those in Experiment 1, except for the amount of electrical stimulation. The level of electrical stimulation in each condition was fixed according to the previously specified criteria. After subjects received five stimuli in each condition, the device was retrieved, and they were asked to respond to the following questions on a 7-point Likert scale regarding the sensation of the stimulation and the perception of the cat:

- Q1: Did the stimulus feel like electrical stimulation? (1: no, 7: yes)
- Q2: Did the stimulus feel like a cat scratch? (1: not like a cat scratch, 7: very much like a cat scratch)
- Q3: How uncomfortable was the stimulus? (1: not uncomfortable, 7: extremely uncomfortable)
- Q4: How intense was the pain? (1: mild, 7: severe)
- Q5: Did it feel as if the cat was present? (1: not at all, 7: very much)
- Q6: Did the stimuli feel as if they matched the sensation of the cat's scratching? (1: does not match, 7: matches exactly)

Since the threshold for discomfort was measured in advance without presenting the cat animation, the participants were instructed to answer Q3 by regarding the value from the condition without the cat animation as 4 (reference).

The data were analyzed using the Friedman test for each question in Experiment 2 to assess whether differences existed among conditions. If differences were identified, a Wilcoxon signed-rank test with Bonferroni correction was applied for each condition.

3.2 Results and Discussion

Figure 6 presents the response values for each question in each condition. The average response rate for the cat liking questionnaire before the experiment was 5.83, and no subject provided a response lower than 5. This suggests that none of the subjects experienced aversion to the cats presented in the images.

Regarding whether the stimulation was perceived as electrical (Q1), significant differences between nothing and cat, and nothing and lag ($p < 0.01$, $p < 0.05$, respectively) was found. This suggests that placing the cat on the smartphone screen made it more difficult to interpret the stimulation as electrical.

Regarding whether the stimulation was perceived as resembling the scratching of a cat (Q2), a significant difference was found between “nothing” and “cat” and between “nothing” and “lag” ($p < 0.01$, $p < 0.05$, respectively). This suggests that the cat animation made subjects to interpret the stimulation as originating from the cat.

In the discomfort level (Q3), a significant difference was observed between nothing and cat ($p < 0.05$). This suggests that displaying an animation of a cat on the screen, synchronized with the electrical stimulation, reduced the discomfort level compared to when no image was displayed.

In the pain intensity (Q4), no significant differences were found between the conditions, suggesting that the perceived pain remained consistent across all conditions.

Regarding whether the cat appeared to be present (Q5), a significant difference was found between “nothing” and “cat” and between “nothing” and “lag” ($p < 0.01$ for both). This suggests that the cat animation led subjects to feel as though the cat was present, compared to when the cat was not displayed.

Regarding the matching of the stimulation (Q6), significant differences were found between nothing and cat, nothing and lag, and cat and lag ($p < 0.01$, $p < 0.05$, and $p < 0.05$, respectively). These results confirm that the subjects were able to distinguish between the conditions between the cat and lag.

In Experiment 2, a reduction in discomfort was observed in the cat condition across all subjects, and the results of Q1 and Q2 confirmed that displaying an animated cat on the smartphone tended to shift the interpretation of the stimulus from electrical to the cat’s scratching stimulus, regardless of whether the animation and electrical stimulus were delayed. This shift may have occurred because subjects perceive a certain degree of reality through the presentation of the cat animation as visual information, as shown in the results of Q5. However, this interpretation was weaker in the lag condition. Subjects also reported that the delay made them feel uncomfortable, as if it were merely an electrical stimulus. More subjects noticed the delay difference between the two conditions than in Experiment 1. This difference can be attributed to the inclusion of a question about the delay in Q6.

Regarding the discomfort level (Q3), a significant difference was found only between the nothing-cat condition, similar to the result of Experiment 1, suggesting that the effect of context modification may be diminished by the inclusion of a delay. On the other hand, the results of Q4 suggest that only the subjective interpretation of pain was altered by context modification, while the perception of stimulus intensity remained unchanged across all conditions. These trends were observed across subjects, unlike in Experiment 1, suggesting that changes in the interpretation of the cause of stimulation due to context modification may occur even in individuals with low electrical tolerance. Some subjects whose baseline for electrical stimulation was below 4 mA reported that condition cat was the least unpleasant, as it felt like a cat scratching them,

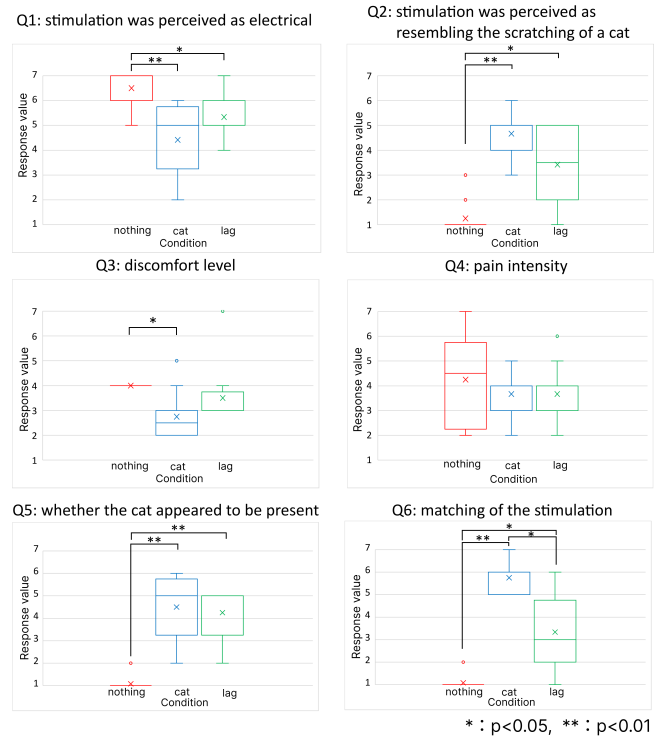


Figure 6: Results for each question in Experiment 2

while others found condition nothing to be more unpleasant than the other conditions.

4 Future Work

In the future, the effectiveness of this method will be evaluated in a VR environment using an HMD to examine its impact on different areas of the body. Additionally, because this method does not distract the user from the source of stimulation, it could be useful for self-treatment tasks such as shaving and hair removal. The method will also be tested in an AR environment, which is more practical for daily use.

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